Exercise 1. (a) Let (X, Y) be uniform on the unit disk given by $D = \{(x,y) \mid x^2+y^2 \le 1\}$.

First, we compute $\mathbb{P}(X \le 1/2 \mid Y = 1/2)$. Observe that

$$\begin{aligned} x^2 + (1/2)^2 &\le 1 \\ x^2 + 1/4 &\le 1 \\ x^2 &\le 3/4 \end{aligned} \implies x \in [-\sqrt{3}/2, \sqrt{3}/2].$$

Thus, X is uniform on the interval $[-\sqrt{3}/2, \sqrt{3}/2]$. By finding the cumulative distribution function of the continuous uniform distribution, we obtain

$$\begin{aligned} a &= -\sqrt{3}/2 \\ b &= \sqrt{3}/2 \\ c &= 1/2 \\ a &\le c \le b \end{aligned}$$

$$\begin{aligned} \mathbb{P}(X \le 1/2 \mid Y = 1/2) &= \frac{c-a}{b-a} \\ &= \frac{1/2 - (-\sqrt{3}/2)}{\sqrt{3}/2 - (-\sqrt{3}/2)} \\ &= \frac{\frac{1+\sqrt{3}}{2}}{\sqrt{3}} \\ &= \frac{3+\sqrt{3}}{6} \approx 0.789. \end{aligned}$$

Next, we evaluate $\mathbb{P}(X \le y \mid Y = y)$ similarly. Since

$$\begin{aligned} x^2 + y^2 &\le 1 \\ x^2 &\le 1 - y^2 \end{aligned} \implies x \in [-\sqrt{1-y^2}, \sqrt{1-y^2}],$$

X is uniform on the interval $[-\sqrt{1-y^2}, \sqrt{1-y^2}]$. Hence, it follows that

$$\mathbb{P}(X \le y \mid Y = y) = \frac{y + \sqrt{1-y^2}}{2\sqrt{1-y^2}}.$$

Therefore, in general,

$$\boxed{\mathbb{P}(X \le Y \mid Y) = \frac{Y + \sqrt{1-Y^2}}{2\sqrt{1-Y^2}}}.$$

$\square$